# Keyboards

A physical keyboard can be an essential input device for entering text, playing games, controlling apps, and more.

People can connect a physical keyboard to any device except Apple Watch. Your users tend to use a physical keyboard all the time and frequently when they are at their desk. Many people prefer using a physical keyboard to the point that they purchase a physical keyboard when entering a UI for work.

Keyboard users often appreciate using keyboard shortcuts to perform tasks in your app. Keyboard shortcuts are a useful shortcut to combination of a primary key and a number of modifier keys (Control, Option, Shift, and Command) that you can use to perform a particular action in your app. For example, Command-Shift-Z is a common shortcut for a redo action.

Apple defines standard keyboard shortcuts to work consistently across all apps, so people can use their existing knowledge and muscle memory. Some apps define custom keyboard shortcuts for the app-specific commands people use most frequently. Keep in mind that people may find it difficult to use the keyboard to control the game, for instance, see Games.

## Best practices

Support keyboard shortcuts whenever possible. Available in iOS, iPadOS and Mac Catalyst, keyboard shortcuts make the people navigate and activate windows, menus, controls, and other interface elements in your app. Use the standard keyboard shortcuts to your app or game, from Cut in UIKit text fields and of the system-supplied settings app. For developer guidance, see Keyboard navigation and shortcuts in Mac Catalyst and UIKeyCommand.

**Important**

Although iPadOS supports keyboard navigation in text fields, it does not support keyboard navigation in other UI elements such as buttons and other controls. Instead, people use Full Keyboard Access to activate controls, navigate to all onscreen components, and perform other actions using the keyboard. For more information, see WWDC 21 for developer guidance, see Focus-based navigation.

Respect standard keyboard shortcuts. While only most apps people generally expect to rely on the standard keyboard shortcuts, people may use a wide variety of keyboard shortcuts in your app offers a unique action that people perform frequently, prefer creating a custom keyboard shortcut. Instead of assigning a custom keyboard shortcut to a standard keyboard shortcut. While playing a game people may expect to use a different standard keyboard shortcuts — such as Command-Q to quit the game. Use the Escape key to pause the game, or Command-Shift-Q to log out. For more guidance, see Games portrait.

## Standard keyboard shortcuts

Apple defines a set of standard keyboard shortcuts for widespread actions. People can and often expect that the shortcuts they know work differently in your app or game. Only consider creating a custom keyboard shortcut for a standard shortcut in your experience. For example, if your app or game doesn't support text editing, it doesn't need a text editing command-like Ctrl, Shift, and Command keys for the action that has never otherwise, like Ctrl-Alt.

People expect each of the following keyboard shortcuts to perform the action listed in the table below.

| Primary key | Keyboard shortcut | Action |
| --- | --- | --- |
| Space | Command Space | Show or hide the Spotlight search field |
| | Shift-Command-Space | Switch keyboard |
| | Control-Command-Space | Show the Spotlight search results and keyboard |
| | Control-Shift-Command-Space | Show the Spotlight Character Viewer |
| Tab | Shift-Tab | Navigate through controls in reverse direction |
| | Command-Tab | Switch to the next app in the app switcher |
| | Shift-Command-Tab | Switch to the previous app in the app switcher |
| Esc | Esc | Cancel the current operation |
| Esc | Option-Command-Esc | Open the Force Quit dialog |
| Esc | Shift-Option-Command-Esc | Quit all open other apps (except Finder and System Preferences) and restart the Mac |
| F1 | Control-F1 | Toggle UI keyboard access on or off |
| F2 | Control-F2 | Move focus to the menu bar |
| F3 | Control-F3 | Move focus to the Dock |
| F4 | Control-F4 | Move focus to the active or next window |
| F5 | Control-Shift-F5 | Move focus to the previously active window |
| F6 | Control-F6 | Move focus to the next pane |
| F7 | Command-F5 | Turn VoiceOver on or off |
| F8 | Command-F6 | Move focus to the previous pane |
| F9 | Control-Shift-F9 | Move focus to the previous control |
| F10 | Control-F9 | Move focus to the next control |
| F11 | Command-F11 | Temporarily conceal all windows to reveal the desktop |
| F12 | Command-F12 | Show Dashboard |
| F13 | Control-F12 | Show Widgets |
| F14 | Control-F13 | Show desktop |
| F15 | Control-F14 | Slide to display Dock and menu bar |
| Grave accent (`) | Command-Grave accent (`) | Move focus to the next window in the front app |
| | Shift-Command-Grave accent (`) | Move focus to the previous window in the front app |
| | Option-Command-Grave accent (`) | Move focus to the next window in the app |
| Hyphen (-) | Command-Hyphen | Decrease the size of the selected item |
| | Option-Command-Hyphen | Zoom out when current window is zoomed |
| Left bracket ([) | Command-Left bracket ([) | Left align a selection |
| Right bracket (]) | Command-Right bracket (]) | Right align a selection |
| Period (.) | Command-Period (.) | Cancel an action |
| Comma (,) | Command-Comma (,) | Display the Spotlight preferences |
| Semicolon (;) | Command-Semicolon (;) | Find misspelled words in the document |
| Comma | Command-Comma | Open the app's preferences window |
| Forward slash (/) | Control-Forward slash (/) | Navigate between text fields |
| Period (.) | Command-Period (.) | Cancel an operation |
| | Shift-Command-Period (.) | Increase screen zoom |
| Apostrophe (') | Command-Apostrophe (') | Open a new window |
| Equal sign (=) | Shift-Command-Equal sign (=) | Turn zoom on |
| | Option-Command-Equal sign (=) | Zoom in when screen zoomed in |
| 0 | Shift-Command-0 | Calibrate the screen for the Clipboard |
| 1 | Shift-Command-1 | Calibrate the screen for the Clipboard |
| 2 | Shift-Command-2 | Calibrate a selection for the Clipboard |
| 3 | Shift-Command-3 | Take a screenshot and save it to a file |
| 4 | Shift-Command-4 | Take a screenshot of a selected area and save it to a file |
| 5 | Shift-Command-5 | Open the screenshot utility |
| 6 | Shift-Command-6 | Open a new Touch Bar window |
| A | Command-A | Select all items |
| B | Command-B | Toggle bold text |
| C | Command-C | Copy the selected item |
| D | Command-D | Duplicate the selected item |
| E | Command-E | Find the selection in the active app |
| F | Command-F | Open a Find window |
| G | Command-G | Find the next occurrence of the found item |
| H | Command-H | Hide the front app's windows |
| I | Command-I | Toggle italic text |
| J | Command-J | Scroll to a selected item |
| K | Command-K | Open a new URL |
| L | Command-L | Open the Location dialog |
| M | Command-M | Minimize all windows of the front app to the Dock |
| N | Command-N | Open a new document |
| O | Command-O | Open a dialog to display the selected item |
| P | Command-P | Open a dialog for printing the current document |
| Q | Command-Q | Quit the app |
| R | Command-R | Refresh the contents of the window |
| S | Command-S | Save the current document |
| T | Command-T | Show or hide the font window |
| U | Command-U | Toggle underline style |
| V | Command-V | Paste the item from the Clipboard |
| W | Command-W | Close the window |
| X | Command-X | Cut the selected item |
| Y | Command-Y | Toggle the Quick Look preview |
| Z | Command-Z | Undo the last command |
| | Shift-Command-Z | Redo the last undo command |
| Left arrow | Command-Left arrow | Move the selected item to the left |
| | Option-Command-Left arrow | Hide other apps |
| | Shift-Command-Left arrow | Move the selected item to the left by a large increment |
| | Control-Command-Left arrow | Move the selected item to the left by a small increment |
| Right arrow | Command-Right arrow | Move the selected item to the right |
| | Option-Command-Right arrow | Show the next app in the app switcher |
| | Shift-Right arrow | Extend selection to the next character in the direction of the text flow |
| | Option-Shift-Right arrow | Extend selection to the end of the next word |
| | Control-Right arrow | Move focus to another window in the same app, such as a dialog |
| Left arrow | Command-Left arrow | Change the keyboard input source to the next source in the list |
| | Shift-Left arrow | Extend selection one character to the left |
| | Shift-Option-Left arrow | Extend selection to the beginning of the current word, especially the end of the current word |
| | Shift-Right arrow | Extend selection to the beginning of the next word |
| | Control-Left arrow | Move focus to another window in the same app, such as a dialog |
| Up arrow | Command-Up arrow | Move the selected item up |
| | Shift-Up arrow | Extend selection to the previous paragraph, especially the beginning of the current paragraph, or the previous paragraph if the current paragraph is selected |
| | Shift-Option-Up arrow | Extend selection to the beginning of the current paragraph, or the previous paragraph if the current paragraph is selected |
| | Control-Up arrow | Move focus to another window in the same app, such as a dialog |
| Down arrow | Shift-Command-Down arrow | Extend selection to the end of the current paragraph, or the next paragraph if the current paragraph is selected |
| | Shift-Down arrow | Extend selection to the next paragraph, especially the end of the current paragraph, or the next paragraph if the current paragraph is selected |
| | Option-Shift-Down arrow | Extend selection to the end of the current paragraph, or the next paragraph if the current paragraph is selected |
| | Control-Down arrow | Move focus to another window in the same app, such as a dialog |
| Forward slash | Shift-Command-Forward slash | Show the Spotlight Help menu |
| Delete | Command-Delete | Move the selected item to the trash |
| | Shift-Command-Delete | Empty the trash |
| | Option-Shift-Command-Delete | Empty the trash without confirmation |
| | Control-Delete | Move focus to the next pane |
| | Shift-Control-Delete | Move focus to the previous pane |
| | Option-Control-Delete | Move focus to the next control |
| | Shift-Option-Control-Delete | Move focus to the previous control |
| | Command-Delete | Move focus to the next window in the front app |
| | Shift-Command-Delete | Move focus to the previous window in the front app |
| | Option-Command-Delete | Move focus to the next window in the app |
| Page Up | Shift-Page Up | Extend selection to the previous page |
| Page Down | Shift-Page Down | Extend selection to the next page |
| Home | Shift-Home | Extend selection to the beginning of the current line |
| End | Shift-End | Extend selection to the end of the current line |
| Tab | Option-Tab | Move focus to the next window in the app |
| Space | Shift-Space | Scroll up a page |
| | Option-Space | Scroll down a page |
| | Control-Space | Change the input source to the next source in the list |
| | Shift-Control-Space | Change the input source to the previous source in the list |
| | Option-Control-Space | Show the Spotlight Help menu |
| | Command-Space | Scroll to a selected item |
| | Shift-Command-Space | Scroll up a page |
| | Option-Command-Space | Scroll down a page |
| | Control-Shift-Command-Space | Scroll to the end of the current document |
| | Control-Option-Command-Space | Scroll to the beginning of the current document |
| Shift | Shift-Command-Shift | Toggle the Finder sidebar |
| | Option-Shift-Command-Shift | Toggle the Finder preview |
| | Control-Shift-Command-Shift | Toggle the Quick Look preview |
| | Shift-Command-Shift-Period (.) | Toggle the Finder hidden files |
| Control | Shift-Control-Shift | Move focus to the next pane |
| | Option-Control-Shift | Move focus to the next control |
| | Command-Shift | Move focus to the next window in the front app |
| | Option-Command-Shift | Move focus to the next window in the app |
| Option | Shift-Option-Shift | Move focus to the next pane |
| | Control-Option-Shift | Move focus to the next control |
| | Command-Option-Shift | Move focus to the next window in the front app |
| | Option-Command-Option-Shift | Move focus to the next window in the app |
| Command | Shift-Command-Shift | Move focus to the next pane |
| | Option-Command-Shift | Move focus to the next control |
| | Control-Command-Shift | Move focus to the next window in the front app |
| | Option-Command-Option-Shift | Move focus to the next window in the app |

The system also defines several keyboard shortcuts for use with Mission Control, Dashboard, Spotlight, Dictation, Keyboard Access, and input methods. These shortcuts are not documented directly in Mains commands.

| Keyboard shortcut | Action |
| --- | --- |
| Control-Shift-Command-Space | Toggle the Dictation input source |
| Control-Option-Space | Switch to the next input source in the list |
| Shift-Mac Keyboard Space | Dictate |
| Command-Shift-Space | Toggle the Dictation input source |
| Command-Shift-Space | Toggle the Dictation input source |

## Custom keyboard shortcuts

Define custom keyboard shortcuts for only the most frequently used actions. Consider creating custom keyboard shortcuts for using keyboard shortcuts for actions people perform frequently, but defining too many new shortcuts can make your app seem difficult to learn.

Use modifier keys in ways that people expect. For example, pressing Command-shift-straight moves items as a group, and Command-shift-diagonal moves items individually. Use the arrow keys to navigate in a direction. Holding an arrow key moves the selected item by the smallest gap defined until it absorbs until people release the key.

Here are the modifier keys and the symbols that represent them:

| Modifier key | Symbol | Recommended use |
| --- | --- | --- |
| Command | ⌘ | Use for most of the keyboard shortcuts. Use for the most common actions. |
| Shift | ⇧ | Modify the Shift key to create a variant of a command. |
| Option | ⌥ | Use the Option key for variants for less common actions. |
| Control | ⌃ | Avoid using the Control key. Use Command, Shift, and Option instead. |

Tip

Some languages require modifier keys to generate certain characters. For example, on a French keyboard, Option-5 produces the left bracket character ([). Adding Shift and Command key as a modifier, but avoid using additional modifier with characters that aren't available on all keyboards. Use the Option key to create variants. Control-Option, Long-Control, with Command.

**List modifier keys in the correct order.** If you use more than one modifier key, list them in the following order: Control, Option, Shift, Command.

**Avoid adding Shift to a shortcut that uses the upper case.** Some keyboard shortcuts include the Shift key, and people expect that they must hold the Shift key to type the uppercase character. If a new character key, as it's clearer to simply list the upper case character. For example, Command-Shift-Z is a common shortcut for Redo Redo for a Command-Shift, where Command-Shift-Z is a common shortcut for Command-Quarter-Hold, not Shift-Command-Z.

**Let the system localize and mirror your keyboard shortcuts as needed.** The system automatically localizes a shortcut's modifier keys and mirrors the shortcut's modifier keys for right-handed. If your app or game switches to a right-to-left layout, the system automatically mirrors the shortcut. For guidelines, see Right-to-left.

**Avoid creating a new shortcut by adding a modifier to an existing shortcut.** Adding a modifier to a shortcut is confusing because people are accustomed to using Command-Z for Undoing an action. It would be confusing to use Shift-Command-Z for Redo instead. Use a different keystroke to create and use.

## Platform considerations

No additional considerations for iOS, iPadOS, macOS, or tvOS. Not supported in watchOS.

## visionOS

visionOS supports a set of keyboard shortcuts appear in the shortcut interface that displays when people hold the Command key on a connected keyboard. The shortcuts are organized into tabs and labels. For guidance, see Keyboard shortcuts.

Apple Vision Pro displays app commands in familiar system-provided interfaces such as menus. For a list of system-provided menu bar menus, the shortcut interface displays all relevant categories in one view, listing within each category only commands that are available in the current context.

**While developing keyboard shortcuts, Xcode and the shortcut interface displays a flat list of all items in each category.** Organize your app's commands into menus, and assign them to tabs. Make sure each shortcut's title is descriptive enough for people to choose without the additional context a scheme. For guidance, see Keyboard shortcuts and Accessibility.

**Recognize that people are an overlay when they use a keyboard shortcut.** Keyboard shortcuts in visionOS, people connect a physical keyboard and use it to control visionOS app or game, the system displays a virtual keyboard overlay that provides typing completion and other controls.

Play

# Resources

## Related

Virtual keyboards

Entering data

Pointing devices

## Developer documentation

`KeyboardShortcut` — SwiftUI

Input events — SwiftUI

Handling key presses made on a physical keyboard — UIKit

Mouse, Keyboard, and Trackpad — AppKit

# Change log

| Date | Changes |
| --- | --- |
| June 9, 2025 | Moved game-specific key bindings guidance to the Game controls page. |
| June 10, 2024 | Added game-specific guidance and made organizational updates. |
| June 21, 2023 | Updated to include guidance for visionOS. |




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